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Review

Advancing Precision Agriculture Through Digital Twins and Smart Farming Technologies: A Review

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Abstract: The agricultural sector is evolving with the adoption of smart farming technologies, where Digital Twins (DTs) offer new possibilities for real-time monitoring, simulation, and decision-making. While previous research has explored the Internet of Things (IoT), UAVs, machine learning (ML), and remote sensing (RS) in enhancing agricultural efficiency, a systematic approach to integrating these technologies within a DTs ecosystem remains underdeveloped. This paper presents a systematic review of 167 studies published between 2018 and 2025. The objective of this study is to examine recent advancements in DTs-enabled precision agriculture and propose a comprehensive framework for designing, integrating, and optimizing DTs in smart farming. The study systematically examines the current state of DT adoption, identifies key barriers, and computational efficiency challenges, and provides a step-by-step methodology for DT implementation. The review sheds light on potential future research direction and implications for policy, with the aim to speed up the adoption of DTs-based farm management systems in their operational success and commercial viability through analysis of practical applications and future perspectives. This study presents an innovative strategy for integrating digital and physical systems into agriculture and is an important contribution to existing literature.

Keywords: digital twins; systematic review; precision agriculture; smart farming; IoT; UAVs; machine learning; remote sensing; data integration; sustainable agriculture

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1. Introduction

The advancement of technology under the Fourth Industrial Revolution (Industry 4.0) is driving the transformation of agriculture into Agriculture 4.0, where traditional farming practices are being enhanced through the integration of Information and Communication Technologies (ICT), the Internet of Things (IoT), Artificial Intelligence (AI), and data analytics [1–4]. The result of these advancements is real-time insights, automation, and precision-based interventions that have drastically improved agronomy and sustainability throughout farms. Within agricultural systems, Digital Twins (DTs) represent the virtual version of the corresponding farm, where real-time data collected from IoT sensors, Unmanned Aerial Vehicles (UAVs), and remote sensing (RS) technologies are integrated to improve decision-making and productivity. Using machine learning and predictive modeling, DTs allow farmers to monitor, simulate, and optimize farming operations. These

systems enhance efficiency and sustainability by providing a continuously updated and interactive virtual model of the farm that exchanges real-time data with its physical counterpart. DTs incorporate IoT sensors, UAVs, RS technologies, and AI-driven analytics to formulate predictive modeling, simulation, and optimization of farming operations [5,6]. Although successful implementations of DTs are known in other industries, such as manufacturing, aerospace, and healthcare [7,8], their use is less developed in this field, where DTs could support decision-making, resource efficiency, and climate adaptation [9–11].

Current research on DTs in agriculture primarily focuses on isolated applications such as irrigation management, disease detection, and crop monitoring, rather than presenting a unified, scalable framework for their deployment [9–11]. Different studies have addressed IoT, UAVs, machine learning (ML), and remote sensing in smart farming, but the concept of DTs is underdeveloped. This review intends to integrate the recent advances in smart farming technologies while proposing an extensive framework for deploying DTs in precision agriculture. Although previous literature focused on the role of IoT, UAVs, ML, and RS in streamlining farm operations, a systematic integration approach to effectively embed these technologies within a DTs ecosystem is missing.

Figure 1 shows the block diagram that illustrates the architecture of a DT system in smart farming. The diagram shows how real-time data is collected from the physical farm using advanced technology such as IoT sensors, UAVs, and other monitoring devices. This data is then transmitted to a virtual representation (Digital Twins), where it is analyzed using AL and ML techniques. The insights generated are used for decision-making, which supports precision interventions in the physical farm, creating a continuous feedback loop between real and digital environments.

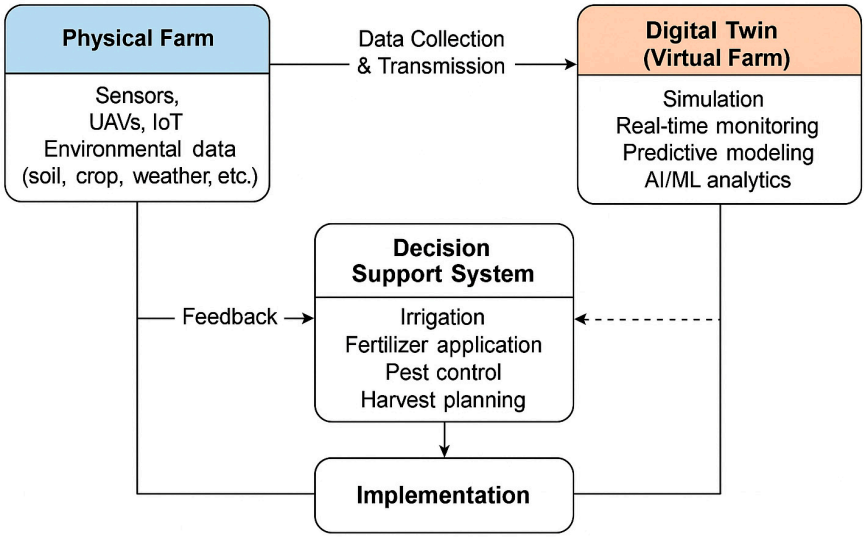


Figure 1. Block Diagram of DTs Architecture in Smart Farming.

Our objective is to address this gap by establishing a clear methodology for designing, integrating, and optimizing DTs in agriculture, thereby enabling the uniform fusion of digital and physical agricultural systems. The specific objectives of this study are: (1) to analyze the current state of DT adoption in agriculture by reviewing recent advancements, case studies, and technological developments, (2) to identify key barriers to DT implementation and explore solutions to challenges related to data integration, computational efficiency, and scalability, (3) to propose a comprehensive DTs framework outlining a step-by-step methodology for designing, integrating, and optimizing DTs in precision agriculture, and (4) to highlight future research directions and policy implications necessary to accelerate the adoption of DTs in smart farming. Furthermore, we propose a conceptual framework

for DTs-driven farm management, illustrating its potential for operational success and commercial viability. A conceptual framework for linking physical and virtual farm systems is illustrated in Figure 2.

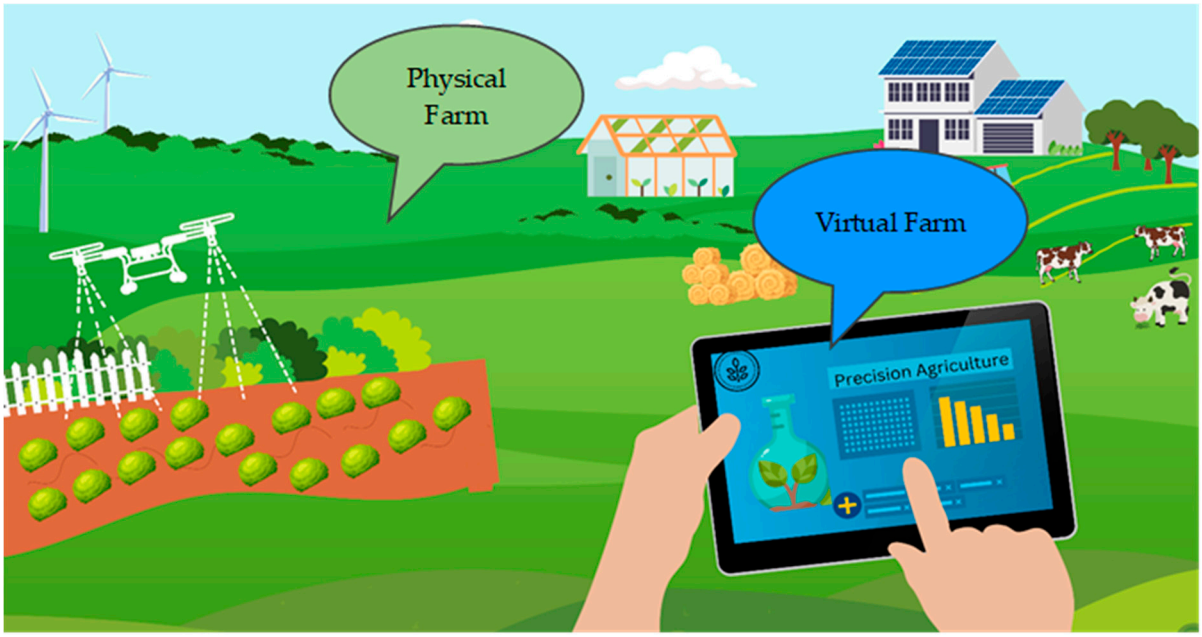


Figure 2. Smart Farming with DTs: Connecting Physical and Virtual Farms.

2. Literature Review: DTs in Agriculture

2.1. Definition of DTs

DTs are a virtual model of a physical object, system, or process that progressively updates and changes over a period as it receives real-time data from the physical version. It combines sensors, artificial intelligence (AI), and data analytics to monitor, simulate, and optimize performance, providing predictive insights and informed decision-making [12]. DTs are not constant, they continuously update with live data to reflect the real-world challenges. This technology is used widely in various industries like agriculture, manufacturing, healthcare, and smart cities to improve efficiency, reduce risks, and enhance the performance of systems. The DTs hold significant importance in agriculture [13]. The conceptual model of DTs comprises three core components: (a) physical objects, indicating what exists in the physical world; (b) virtual objects, which refer to the representations of the physical counterpart in a virtual setting; and (c) the connecting data and information network that interrelates virtual and physical entities. This model includes the creation of a digital replica of real-world assets like farms and crops. Farmers can use this technology to monitor and evaluate real-time data regarding plant growth, soil health, weather conditions, and their fields. The concept of DTs can provide a successful framework for implementing Cyber-Physical Systems (CPS), which can be utilized in multiple fields of the industry, including product design, production, supply chain, and agriculture [14]. The DT approach empowers agriculture and agricultural practices with more efficiency, precision, and sustainability, resulting in better crop yields and resource management. DTs can significantly reshape plant production and reduce the harmful effects on plants by observing the specific properties of soil. Moreover, DTs help optimize water usage, and healthcare, and also reduce dependency on chemical fertilizers, promoting sustainable farming practices. Figure 3 illustrates the concept of DTs in agriculture.

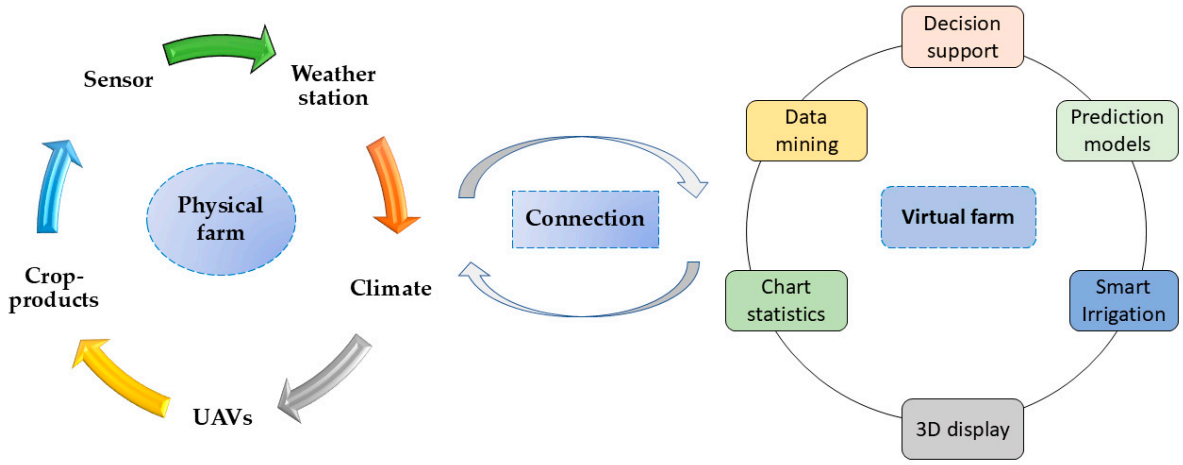


Figure 3. A simplified view of the data interaction between physical and virtual farms [12].

Therefore, based on the DTs model, a combination of machine learning models, big data analysis, and decision support systems provides continuous monitoring and analysis of the physical environmental conditions, soil components, and irrigation techniques of agricultural land. This holistic approach is important for maximizing resource use and developing sustainable agricultural practices [15]. These DTs are a valuable means for optimizing decision-making, allowing for simulating the effects of environmental and genetic factors on animal performance and breed production. This helps to minimize harmful effects on the environment while optimizing productivity. Moreover, the constant interchange and analysis of data are possible by connecting DTs with several other digital platforms, such as supplier databases, veterinary data, and farm management systems. Such a capability helps in quality assurance, traceability, and better decision-making processes in the complete value chain. A possible strategy could involve improving plant health and production efficiency, supported continuously by literature on the development of sustainable agricultural practices. Research and development in this area will need to continue to take advantage of digital technologies in modern agriculture effectively and address the associated constraints.

2.2. Fundamental Principle of DTs in Agriculture

A complete digital twin refers to the principle of developing a virtual twin of a real agricultural system. A DT is executed in real-time by the real-world farming context considering the tasks, routines, and practices [16,17]. Data from various sources, such as sensors, satellite imagery, and field observations, provide accurate and comprehensive representations of agricultural systems. However, DTs depend on real-time data feeds from sensors, remote sensing devices, and other data sources. To provide the model with updated segmentation between grain types and their shortages, this data is regularly collected to reflect the state of the agricultural system and adapt the model's parameters. Prior to performing tasks or making decisions using DTs, all data must be verified and integrated in real-time to ensure the accuracy of the DTs and to allow decision-makers to access detailed information [18,19]. This helps farmers and other relevant stakeholders make decisions about irrigation, fertilization, pest control, and all sorts of management practices. DTs provide a visual representation of the agricultural system, allowing users to actively monitor and analyze its various components in real time. This allows for better decision-making due to a complete understanding of the system's current status and performance. The use of data mining and simulation methods progresses the implementation of optimization and decision support goals by allowing for optimum management methods to be identified [20,21].

Understanding the different stages in the creation of a DT is critical, with modeling being one of the most common components of any DT system as it represents and captures the real-world agricultural processes and the detailed underlying structures [22]. There are various approaches to creating these models. A common approach is the physics-based approach, which simulates realistic processes based on mathematical equations and physical laws (e.g., crop growth, soil moisture, water flow in an irrigation system) [23]. Another method is the data-driven approach, where models are built using historical data and machine learning algorithms [24]. These models are capable of learning from sensor data and predicting yield, disease risk, or water needs [25]. A third approach is a hybrid model, which tries to combine both physical understanding and data in order to make the simulation more accurate [26].

These models are built and run using different software tools and platforms. Popular software tools used in agriculture include MATLAB/Simulink (for system modeling and control) [27], Unity3D (for 3D visualization and some interactive environments), AnyLogic (for simulation modeling), Thingworx (for industrial IoT and digital twin integration), and many more [28]. The choice of software depends on the goal of the DTs and the level of complexity required. The level of simulation can also vary. Some DTs are applied for real-time monitoring and decision-making where the model needs to be updated continuously from sensor data. Others are set up for scenario analyses, wherein the user runs simulations to assess "what-if" conditions, for example projecting future crop performance under different irrigation schemes. How to balance the right modeling method, tool, and level of simulation depends on the goals, data available, and the infrastructure of the agricultural system.

In recent years, the terms Digital Model (DM), Digital Shadow (DS), and Digital Twin (DT) have been used in the literature and are sometimes interchangeable. However, these terms describe different levels of systems interaction and data integration. Table 1 shows the comparison of these three concepts, based on their characteristics, such as functionality, data flow direction, and decision-making capability. The definition and distinctions are supported by recent literature to ensure conceptual clarity and accuracy [13,29–32].

Table 1. Digital Twin Defined by Data Integration Levels.

Integration Level	Definition	Key Characteristics	References
Model (Digital Model)	A static digital representation without real-time data flow.	• No automatic updates. • Function as a standalone model.	• Kritzinger et al. (2018) [29]
Partially Integrated (Digital Shadow)	A digital representation with one-way data flow from the physical system to the digital.	• Real-time monitoring. • No feedback/control over the physical system.	• Kritzinger et al. (2018) [29]; AboElHassan et al. (2023) [30]; El Saddik et al. (2018) [9]
Fully Integrated (Digital Twin)	A real-time, bidirectional link between physical and virtual entities.	• Enable simulation, optimization, and decision.	• Kritzinger et al. (2018) [29]; Grieves (2014) [31]; Pylianidis et al. (2021) [13]

As illustrated in Table 1, a Digital Model is a static representation that is not interactive in real time, while a Digital Shadow is characterized by a one-way data flow from the physical environment to the digital environment. The DTs are the most advanced level, where two-way connectivity allows for continuous exchange of data and decision support capabilities. The differences between these setups are an integral part of understanding how

integrated they are, or the level of functionality when these two categories of technologies are used with smart farming systems.

3. Data and Methods

3.1. Design and Search Strategy

The objective of this study is to conduct a comprehensive review of the integration of DT technology in the agricultural sector, exploring its synergy with advanced innovations such as AI, IoT, blockchain, and other emerging technologies. This study will highlight the application areas of DTs in various domains of smart agriculture, providing insights into the current adoption status, implementation challenges, and the transformative potential of Digital Twin systems in enhancing decision-making, resource optimization, and sustainable agricultural practices. The review aims to address the following research questions: (1) How do DTs contribute to precision agriculture by integrating IoT, UAVs, and AI-driven analytics? (2) What are the key enablers and barriers to the adoption of Digital Twin technology in sustainable agricultural practices? (3) How can a conceptual framework for Digital Twin modeling be developed to optimize smart farming operations, and (4) What role does Digital Twin technology play in improving climate resilience, Greenhouse gas (GHG) emissions reduction, and sustainable food production within the context of climate-smart agriculture? This review follows the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines, providing a structured and rigorous approach to reviewing and synthesizing existing literature on Digital Twin technology in agriculture [33].

To collect relevant data, publications from scientific conferences, international journals, and online sources were identified and analyzed. SCOPUS was used as the primary database for literature retrieval, employing the search query: ("Digital Twins in Agriculture") AND ("IoT" OR "UAV"). A total of 261 research articles were initially retrieved from SCOPUS. These articles then underwent a systematic screening process to refine the dataset and ensure the inclusion of high-quality and relevant studies. The first stage of screening involved reviewing titles and abstracts, leading to the removal of 84 studies due to duplication, lack of full text, or irrelevance to the research topic. The remaining 177 studies were subject to full-text screening, resulting in a further 10 studies being excluded based on abstract screening for not meeting the predefined inclusion criteria.

The 167 full-text articles were verified for eligibility based on the research focus on DTs in agriculture. From this process, 165 studies were identified as appropriate for qualitative synthesis (i.e., they could be fully analyzed and included in the systematic review). Such a systematic and logical approach guarantees comprehensive, credible, and relevant literature in the final selection, covering relevant aspects of how DTs, IoT, and UAV technologies play a role in the evolution of precision agriculture. Besides the main list of reviewed studies, we also highlighted several important and recently published papers in Table 2. These papers were not added from outside of the main search. Instead, they were chosen to give readers examples of important research and current developments in the field. The selected studies show how different digital tools such as IoT, UAVs, AI, and other smart technologies are being applied in real farming situations. They cover a variety of uses and demonstrate the value, flexibility, and innovation of DT systems in supporting modern agricultural practices. The selection process started with reviewing the titles and abstracts, followed by reading the full articles to make sure they matched the goals of the study. The final literature search was completed on 5 March 2025. Figure 4 shows the steps we followed using the PRISMA method for identifying, reviewing, and selecting the studies.

Table 2. Important Journal Papers for Literature Identification.

Title	Year	Key Findings/Contributions	References
Instrumentation and Measurement Systems: Digital Twin for Horticulture Farm: Concept and Requirements	2025	The study confirmed that using sensors and digital models in a Digital Twin system can improve monitoring and decision-making on horticulture farms	Postolache et al. (2025) [34]
An Intelligent and Trust-Enabled Farming System with Blockchain and DTs on Mobile Edge Computing	2025	The study confirmed that combining Digital Twins with blockchain and mobile edge computing can improve trust, security, and efficiency in smart farming systems	Rathee G et al. (2025) [35]
DTs: A scientometric investigation into current progress and future directions	2022	The study confirmed growing interest in Digital Twins research and identified key trends and gaps, helping to guide future studies in agriculture and other fields	Kaur H et al. (2025) [36]
Unmanned aerial system and machine learning-driven Digital-Twin framework for in-season cotton growth forecasting	2025	The study confirmed that combining UAV data with machine learning in a Digital Twin framework can accurately forecast cotton growth during the season	Pal P et al. (2025) [37]
A Concentration Prediction-Based Crop Digital Twin Using Nutrient Co-Existence and Composition in Regression Algorithms	2024	The study confirmed that using regression algorithms based on nutrient co-existence can improve the accuracy of crop Digital Twin models for nutrient concentration prediction	Ghazvini A et al. (2024) [38]
Advancing Sustainable Cyber-Physical System Development with a DTs and Language Engineering Approach: Smart Greenhouse Applications	2024	The study confirmed that using Digital Twins with cyber-physical systems and language engineering improves automation and sustainability in smart greenhouse management	Subahi AF et al. (2024) [39]
Novel intelligent grazing strategy based on remote sensing, herd perception, and UAVs monitoring	2024	The study confirmed that integrating UAVs, remote sensing, and herd perception into a Digital Twins system can optimize grazing strategies and livestock management	Chen T et al. (2024) [40]
A review of the current status and common key technologies for agricultural field robots	2024	The study confirmed that technologies like GPS, sensors, AI, and Digital Twins are key to improving the performance and autonomy of agricultural field robots	Liu L et al. (2024) [41]
Model-Driven Development Towards Distributed Intelligent Systems	2024	The study confirmed that model-driven development supports building scalable and intelligent Digital Twin systems for agriculture and other distributed environments	A Barriga (2024) [42]
Internet of robotic things with a local LoRa network for teleoperation of an agricultural mobile robot using a digital shadow	2024	The study confirmed that using a local LoRa network with a digital shadow improves real-time teleoperation and connectivity of agricultural robots in remote areas	Shamshiri RR et al. (2024) [43]
Characterizing the Role of Geospatial Science in DTs	2024	The study confirmed that geospatial science plays a key role in Digital Twins by enabling accurate mapping, monitoring, and spatial analysis in agricultural systems	Metcalfe J et al. (2024) [44]

Table 2. *Cont.*

Title	Year	Key Findings/Contributions	References
Modeling for sustainable groundwater management: Interdependence and potential complementarity of process-based, data-driven and system dynamics approaches	2024	The study confirmed that combining process-based, data-driven, and system dynamics models can improve Digital Twin frameworks for sustainable groundwater management	Secci D et al. (2024) [45]
Potential of Satellite-Airborne Sensing Technologies for Agriculture 4.0 and Climate-Resilient: A Review	2024	The study confirmed that integrating satellite and airborne sensing technologies supports Climate-Resilient Agriculture and strengthens Digital Twin applications in Agriculture 4.0	Hazmy AL et al. (2024) [46]
Research on the Smart Broad Bean Harvesting System and the Self-Adaptive Control Method Based on CPS Technologies	2024	The study confirmed that using cyber-physical systems with self-adaptive control improves efficiency and precision in smart broad bean harvesting	Wang W et al. (2024) [47]

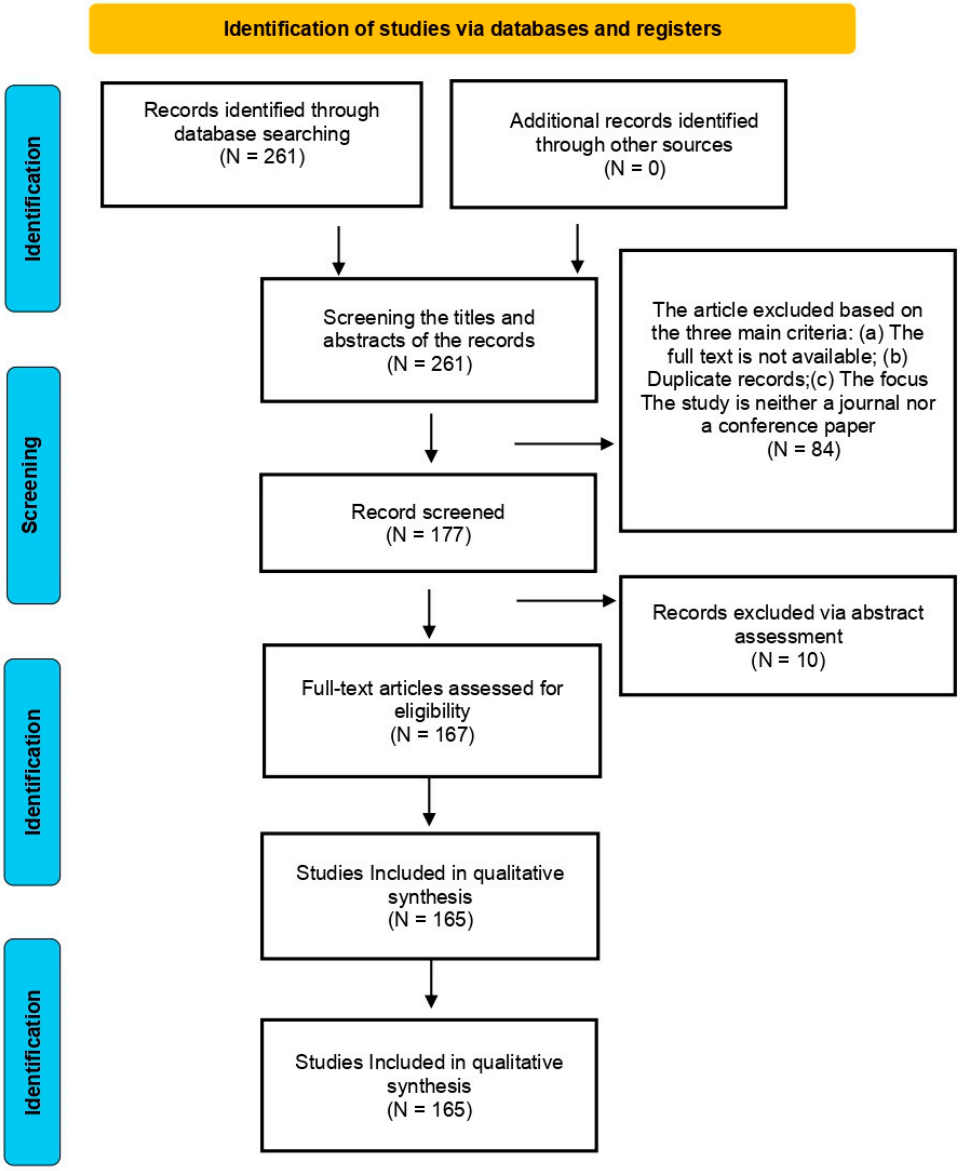


Figure 4. PRISMA flowchart of selected papers for the systematic review.

3.2. Selection Criteria and Quality Assessment

After identifying the relevant documents and excluding duplicates, relevant papers were screened, and low-quality studies or those not directly applicable to the research aims being pursued were excluded from the final dataset. Such criteria were established to allow a focused review of the cases of applications of DTs in agriculture. To be eligible, papers needed to be peer-reviewed journal articles, research reports, research articles, conference papers, or scientific reviews. Only English-language publications were selected to maintain consistency in analysis. A key criterion was that the primary focus of the paper must be on agricultural applications of DT technology, rather than simply mentioning agriculture in passing or as a minor use case.

Many retrieved documents referred to agriculture in a general context or as an example within broader discussions on Digital Twin implementation. Such papers were excluded to ensure that the final document set provided a comprehensive and in-depth understanding of how DT technology is being utilized and advanced within the agricultural sector. Interest in DT technology has greatly increased in the past five years across both academia and industry, accompanied by a growth in the number of related publications, processes, concepts, and envisaged benefits.

Figure 5 illustrates this trend, showing a steady increase in research publications from 2018 to 2025. In the early years (2018–2020), publication numbers were relatively low, reflecting DTs emerging status. However, from 2021 onward, there was a notable rise, indicating expanding research and real-world applications in fields such as agriculture, manufacturing, and healthcare. The sharp rise in publications in recent years shows that researchers and industries are recognizing the potential of DTs to improve efficiency, there are still very few review papers that summarize the existing work and offer a clear framework or direction for future studies. This review brings together the most recent developments, identifies key challenges and research gaps, and provides a useful guide for future research on DTs in agriculture. Overall, Figure 5 highlights the rapid growth of Digital Twin technology, reinforcing its role as a transformative tool in modern industries, with significant potential for optimization, predictive analytics, and real-time decision-making.

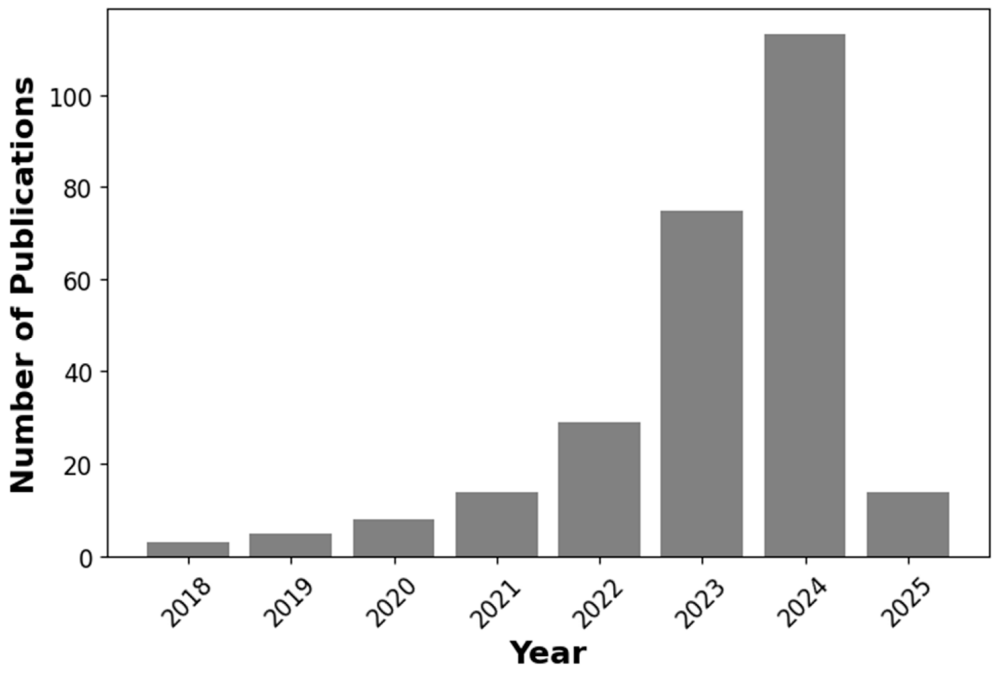


Figure 5. Number of Digital Twin-related publications by the year 2018 to 2025.

4. Digital Twin-Based Smart Farming Technologies

4.1. IoT Framework, UAVs, and DTs Smart Farming

The IoT plays a critical role in DT-based smart farming by enabling seamless data collection, transmission, and analysis. IoT-enabled DTs aggregate data from various sources, such as soil moisture sensors, climate monitoring stations, and UAV-based imaging systems, to enhance precision agriculture. These data-driven insights allow farmers to make informed decisions about irrigation, fertilization, and crop protection strategies. UAVs with different sensors may collect information in the visible, near-infrared, and thermal spectrums, providing valuable insights into crop health, water stress, nutrient deficits, and pest infestations [48,49]. By integrating real-time information from weather stations, soil sensors, and UAV imagery, these models support resource management, irrigation scheduling, and predictive decision-making [50]. Additionally, decision support systems (DSS) powered by DTs provide farmers with automated recommendations for crop management based on AI-driven insights. In examining existing literature, seven specific ways were found in UAVs, which were combined with DTs modeling in the industry, as outlined in Table 3.

Table 3. Application areas of smart farming integration with other DTs.

Application Area	Example from Articles	References
Data management	Collecting real-time data	[51–55]
	Streaming Real-time video	
Tracking progress	Monitoring	[56–61]
	Tracking material on construction sites	
Transportation operations	Earthwork volume computation	[62–66]
	Planning for heavy equipment	
Construction programs	Virtual site visit	[67–71]
	Inspection training simulation	
Structural inspection	Building inspection	[72–75]
	Post-earthquake inspection	
	Bridge inspection	
Construction Safety	Identification of potential hazards Construction safety inspection	[73,75–78]

4.2. Real-World Applications of DTs in Farming

Building on the foundation of IoT-enabled DT systems in smart farming, real-world applications demonstrate how these technologies improve precision agriculture. The application of DTs, by bringing together data from UAVs, IoT sensors, and AI-powered analytics, offers insights that drive actionable decisions to ensure improved monitoring of crop health, optimized irrigation, and biomass management. Below are the subsections that cover essential real-world implementations of DTs in the field of contemporary agriculture, how they are contributing to orchard management, and working on optimizing biomass and resource allocation

4.2.1. Orchard Management Using UAVs

Drone technology, when combined with DTs, has transformed the nature of orchards with real-time monitoring, predictive analytics, and optimization of resources. Conventional orchard management is mainly based on manual work and visual methods, which

are time-consuming and insufficient. Yet, the systematic collection of this information has remained unavailable for managers, even with the use of UAVs, which are equipped with multi-spectral and thermal sensors as well as LiDAR for high-precision, autonomous data acquisition to be further processed using DT models, leading to smarter decision-making and improved orchard management practices.

Zarembo et al. (2023) showed that UAV-based DT models could improve yield estimation and disease detection. Through high-resolution imagery obtained from UAVs, processed using AI-driven analytics, farmers can create digital models of their entire orchards that enable data-driven decision-making. The study examined apples, pears, and cherries, revealing how drones assist farmers in monitoring the health of their crops, identifying stress conditions, and more accurately predicting changes in yield than traditional methods. UAV-based remote sensing immediately detects infections and damage, which is critical to preventing crop loss with high fruit quality [79].

The key benefits of UAV use in orchards include the potential to cover large areas quickly. UAVs with multi-spectral and thermal sensors can cover thousands of acres of orchards while collecting data, including tree health, soil moisture levels, pest infestations, and nutrient deficiencies. Data streams are being compiled into DT models to enable real-time monitoring of orchard conditions. The farmers can accordingly analyze the models and gain a perspective on seasonal variations, weather effects, and growth metrics to make smarter data-driven, economically sustainable decisions by continuously updating such models. A key innovation in the field of orchard management, leveraging UAVs and DTs, is the creation of tree-level digital models.

Pylianidis et al. (2021) analyze 28 case studies related to DT applications in the agricultural sector, mentioning in detail the project called "Digital-Twin Orchard". This initiative employs 3D spinning cameras attached to UAVs to generate DTs for each individual tree, which are then used by farmers to dynamically monitor the health of the tree, devise efficient pruning strategies, and anticipate disease outbreaks. Using tree-specific data through this context, farmers can perform targeted actions, minimizing unnecessary application of pesticides and thus promoting better orchard health [13]. In addition to early foundation reviews such as Pylianidis et al. (2021), recent studies have demonstrated the practical implementation of DTs in agriculture. For example, Rathee et al. (2025) developed a trust-enabled DT system using blockchain and edge computing for smart farming. This model combines DTs with mobile edge computing and blockchain technology to improve trust, traceability, and secure data exchange in agricultural environments. Their system is especially useful for large-scale farms or collaborative agricultural operations where multiple stakeholders need to access real-time data securely [35]. Subahi et al. (2024) proposed a DTs framework for greenhouse automation, integrating CPS and language engineering approaches. This model collected real-time data from sensors that monitored temperature, humidity, and lighting inside the greenhouse. It then simulated environmental conditions and helped automate decisions related to climate control and irrigation. This led to improved productivity, energy efficiency, and sustainable greenhouse operations [39]. Pal et al. (2025) applied UAVs and machine learning in a DTs-based model to forecast cotton growth [37], highlighting the growing integration of AI, IoT, and remote sensing into modern agricultural decision systems. Together, these examples reflect the diverse applications of DTs in agriculture from open-field crops to controlled environments. They show how DT systems can combine technologies like AI, IoT, UAVs, blockchain, and CPS to support precision farming practices and improve agricultural outcomes.

UAVs are also helping with precision irrigation in orchards. Digital Twin models can achieve irrigation scheduling and deliver water distribution by integrating real-time weather data, soil moisture content, and tree transpiration rates. Thermal imaging cameras

on UAVs help detect areas reaching water stress, which enables farmers to adjust irrigation levels in real time. By only using the required amount of water, this technique helps conserve water, which is a benefit for the environment and also reduces costs. Furthermore, UAVs are increasingly utilized for automated pest and disease management in orchards. For example, UAVs with hyperspectral sensors can detect early signs of a disease or pest infestation that may not be visible to the human eye. DT models use this data to predict how an infestation spreads and allow preventative or targeted treatments to be applied. This predictive ability allows for preventive actions by farmers rather than reactive spraying of pesticides, which is beneficial to environmentally friendly agricultural practices. Another important application of DTs in orchards is yield estimation and forecasting, particularly with the use of UAVs. UAVs give high-precision crop forecasts by capturing biophysical parameters like leaf chlorophyll content, fruit size, and canopy density. These insights allow farmers to schedule harvesting, efficiently allocate labor, and forecast supply trends in the market. Through the analysis of past and microdata from real-time UAV operations, Digital Twin models enable tactical predictions and facilitate strategic planning.

In addition, advancements in AI and ML algorithms have also improved the functionality of UAV-DTs in orchards. Automated recommendations for pruning, fertilization, and pest management can be generated from AI-driven models based on the massive amounts of data collected by UAVs [80]. This data-driven strategy eliminates wasteful resource consumption while maximizing farms' bottom lines through data-informed decision-making. The various sensors and types of drones and their capacity are shown in Table 4.

Table 4. Drone varieties and performance features.

Type	Weight (kg)	Function	Capacity
Large	More than 150	These drones are well-suited for pesticide application.	Covers more than 100 acres of farmland.
Medium	25 < weight < 150	Drones can carry thermal cameras, higher-resolution RGB cameras, LiDAR sensors, and multi-spectral sensors for comprehensive data collection	These drones operate at a flight altitude of 50 m and can cover up to 100 acres
Minor	2.< weight < 25	Drones equipped with high-resolution RGB cameras, thermal cameras, hyperspectral cameras, and multi-spectral sensors are valuable for diverse agricultural tasks.	It mostly depends on flight height and covers up to 10–20 acres.
Micro	0.20 < weight < 2	These drones are designed with lightweight multi-spectral cameras, small RGB cameras, and LiDAR sensors for more specialized uses.	When flying at an altitude lower than 100 m, they can cover approximately 5–6 acres.
Sonic	below 0.20 kg by weight	Until now, these drones have not found application in agriculture.	N/A

4.2.2. Biomass Management and Crop Growth Optimization

DTs are revolutionizing biomass management and crop growth, providing agriculture precision-based approaches for monitoring, decision-making, and resource optimization [81]. With IoT networks, UAV imagery, AI-driven analysis, and real-time feeds of environmental data, DT technology allows more accurate assessments of crop health, irrigation, and predictive modeling for yield estimation.

Angin et al. (2020) developed a DT framework for farmland to monitor plants and make decisions. Here, a system of low-power Internet device wireless sensor networks

based on LoRaWAN, integrated with UAV-collected imagery, is used to send real-time environmental data and improve the sensitivity of crop monitoring. At the heart of this DTs ecosystem is a deep-learning model to identify plant diseases and weeds to facilitate early intervention and targeted resource management. The framework has a major strength in its integrated nature, allowing the addition of new data sources and increased modeling functionalities for agriculture solutions [81].

In a different study by Alves et al. (2019), an IoT-based Digital Twin system focused on optimizing crop irrigation to enhance yield. This methodology leverages several data sources, integrating in-field sensors with remote-sensing devices, including weather stations. It provides high-level significance through cloud-based data integration, creating a comprehensive, real-time decision-support tool for farmers. This study highlights the merits of Cloud-Based DTs to offer precise, sustainable, scalable, and accessible solutions for precision irrigation and sustainable farming [15].

Ghandar et al. (2021) explored urban farming decision support built on an agent-based Digital Twin model with an emphasis on aquaponic production systems. The researchers constructed a predictive framework utilizing IoT networks, machine learning models, and data-driven simulations to maximize the generalization of the optimization of controlled crop and fish production. They showed that model-based simulations could perform even better than machine learning-based models, especially when ample amounts of training data are not available, providing a scalable approach for managing urban farms [82].

5. Conceptual Framework for DTs Modeling in Smart Farming

Smart farming has been greatly transformed by DTs and UAV implementations to solve key agricultural problems, including soil state, climate variability, resource allocation, and management efficiency. This can be done using Internet of Things-based devices that can create DTs of physical farm environments, enabling farmers to oversee, create theories from all data, and conduct real-time optimization of agricultural operations across the entire farm using DTs. Smart farming is moving toward precision agriculture, made possible by the integration of various technologies such as IoT networks, cloud computing, big data analytics, machine learning, robotics, and UAV technology [83].

5.1. Integration of UAVs and DTs in Smart Farming

The complementary nature of UAVs and DTs supports advanced farm monitoring, versatile decision-making, and automation, which greatly improves efficient and sustainable farms. UAVs are indispensable for data collection, capturing vital details related to crop health, soil content, pest infestations, and environmental conditions from a bird's-eye view. The collected data is then processed and integrated into the DT architecture, contributing to the digital simulation of the farm environment. The dynamic virtual farmland model, which is updated regularly through actual sensor data, delivers unique insight on-field to farmers. An integrated data-driven framework backbone builds upon the DTs layers, representing different aspects of the digital and physical farm ecosystems. These technologies work in synergy to deliver real-world data via UAV imagery, soil instruments, and IoT devices: (a) A sensor-based physical layer that actively monitors the farm. (b) An AI engine processes this data with feature extraction and fusion to create intelligent insights. The macro adjustment includes an application layer of the system, allowing farmers to visualize environmental monitoring and intelligent decision-making to improve production efficiency and resource utilization.

Access to the physical world of farming and all its associated noise is enabled through a cyber-physical convergence layer (or cyber-physical system), which enables a two-way flow of information between the real farm and its digital twin, facilitating dynamic feedback

loops and automated activities on the farm. It provides an interactive feedback system that improves operational efficiency, enables intelligent decision-making, and simplifies the farm management process [84]. The combination of UAVs with DTs in modern smart farming allows the surveillance of farms in real-time and on a large scale, which can save labor and improve operational efficiency. Compared with conventional monitoring techniques, UAVs with multi-spectral and thermal sensors can deliver high spatiotemporal resolution aerial images of the field to identify plant stress factors, including pests and nutrient deficiencies. They allow for spatial and temporal variations to be captured across vast agricultural landscapes, guaranteeing that strategies in farm management are maximized for the improvement of yield [85].

For example, UAVs are integral to automated mapping and georeferencing, delivering accurate field mapping that aids autonomous farm machinery navigation. Farmers can use DTs constructed from UAV-derived data in order to create a detailed digital reproduction with the ability to visualize the development of crops, estimate the likelihood of danger, and suggest optimal farming methods responding to real conditions. As a result, the real-time simulation capabilities of the DTs mean farmers can test a range of scenarios to see how their crops would perform under changing environmental conditions, pest outbreaks, or soil nutrient levels. By predicting crop yields, resource allocation becomes more efficient and sustainable, as farmers can tailor their inputs accordingly, whether it be fertilizer, pesticides, or planting density. As a result, UAV-empowered DTs support remote on-agricultural tasks where the decision-maker can monitor farm conditions without being physically present, thus supporting enhanced smart farming accessibility, scalability, and resilience.

5.2. Key Features of the Digital Twin Framework

As DTs-based smart farm systems diversified and adapted to various agricultural domains, researchers extended the core features of these systems to keep pace with modern agricultural demands. Connectivity is a core element of DTs, facilitating the automated collection and synchronization between the physical environment and its virtual counterparts across farm entities (technologies, sensors, machines, humans, etc.) and operations. IoT sensors, cloud computing, and edge computing will help to process tons of farm data in real time for accurate and quick farm management.

Another major feature of AgriTech is optimization, which lowers operational expenses, reduces resource waste, and enhances yield forecasts, as well as farm production. Transparency comes from combining the data across expertise and translating those insights into actionable intelligence for farmers. DTs enhance it through back-end geospatial analysis; you can use historical farming data and AI-driven recommendations. Thus, DTs build interactive dashboards that will provide the farmers with visual analytics so that they can maintain easy-to-understand analytics. The proactivity of the DTs supports predictive analytics and automated decision-making, thus reducing risks and preventing losses from occurring. This flexibility in the design of the technology used in DTs-based farming systems enables its versatility, which supports the growth of precision farming from large-scale farms to smaller and more diverse locations. A further important element that is part of the DTs-based components is the interaction of a cyber-physical network, which means the possibility of interaction between the digital model of the farm and the physical systems in agriculture. Based on UAVs, IoT-enabled, and remote sensing data acquisition and processing, cyber-physical integration is performed by synchronizing data into a centralized DTs system in real time.

This allows the digital farm model to be updated in near real-time, facilitating support for automated decision-making, as well as being able to plan targeted measures. Virtual farm models allow farmers to simulate crop growth cycles, try different soil management

techniques, or optimize planting schedules without adjusting to the physical environment. Moreover, DTs offer an opportunity for collaborative farm management where different stakeholders, including farmers, agronomists, and policymakers, can programmatically gain access to shared farm data. Such an analytics-based best practice management can bring a revolution to precision farming owing to the open shared data approach.

Most importantly, the fusion of UAV imaging, AI-based decision support systems, and automated farm machinery will define the extremely autonomous farming solution, where farm operations like seeding, irrigation, fertilization, and harvesting can be controlled automatically through DTs. This real-time fusion of imagery, sensor data, and AI-powered analytics enables precision in crop management, soil health assessment, and farm resource allocation. This interactive feedback loop between the digital and physical farm supports continuous learning and adaptive management, which makes DTs an asset in promoting smart farming solutions. In addition, DTs play a critical role in sustainable agriculture, minimizing the unnecessary application of fertilizers and pesticides to secure favorable environmental impacts and continuous productivity. The underlying framework, presented in Figure 6, shows that DTs enable smart agricultural decision-making, modeling, and enhanced adaptability of the farm systems through continuous data feedback, which improves all processes involved.

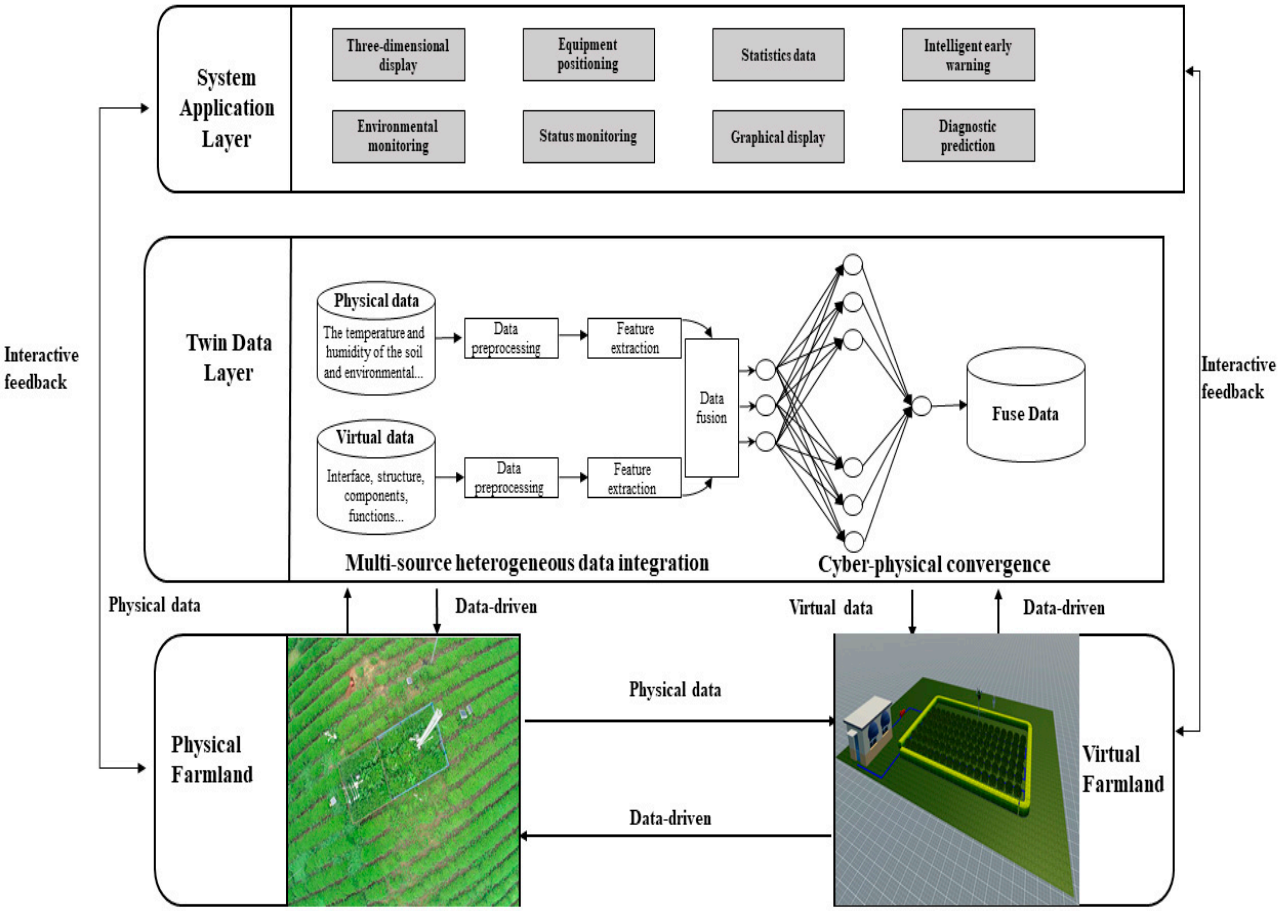


Figure 6. Conceptual diagram for DT technology in smart farming.

6. Discussions

This section is structured to answer four critical research questions suggested at the beginning of this review. Section 6.1 highlights the importance of further developing DTs and smart farming technologies in order to increase both efficiency and productivity in agriculture. Section 6.2 highlights the core barriers to the adoption of DTs and discusses

possible solutions and policies to instigate their utilization in smart farming. In Section 6.3, we provide an in-depth analysis of this framework for DTs and consider some practical aspects, such as how a well-designed system can enhance the optimization of resource utilization, improve the accuracy of yield predictions, and support sustainable agricultural practices. In the end, Section 6.4 outlines potential areas for future research and highlights the implications of these findings for policymaking, calling for further innovation and planning to leverage the benefits of digital technologies in agriculture. Overall, these topics serve as a roadmap for DT implementation, showcasing the potential of these technologies to transform contemporary agriculture, streamline resource utilization, and facilitate data-informed decision-making.

6.1. Advancements Needed in DTs and Smart Farming Integration

A summary of existing technological gaps, quality improvement, accuracy, and real-time decision-making in DT applications alongside smart farming will be discussed in this section. DTs have the potential to revolutionize precision agriculture by enhancing decision-making, increasing productivity, and optimizing resource use [86]. However, several challenges hinder their full-scale implementation. One major issue is data interoperability, which is a big problem with agriculture technology devices, which are built or delivered in so many different formats, making it impossible to combine the data for analysis [87]. Computational efficiency and real-time processing are major issues because DTs involve large-scale data collection and rapid analysis to deliver timely insights for farmers [88,89]. Furthermore, sensor reliability also plays an important role because DTs rely on time-derivative data from IoT devices, UAVs, and remote sensing systems, which are prone to errors, environmental interference, or connectivity issues [90,91]. Another critical challenge is cybersecurity, as DTs deal with large amounts of sensitive agricultural data, making them sensitive to cyber threats and unauthorized access [92–94]. To fully harness the potential of DTs in agriculture, research should center its attention on scalable, efficient, and secure DTs applications by improving AI-based analytics, creating standardized protocols for data exchange, and enhancing cyber-physical integration [95]. By tackling these technological deficiencies, DTs can serve as a reliable and viable option for smart farming, aiding farmers in better decision-making and promoting sustainable agricultural practices.

Computational efficiency and real-time processing limitations are major challenges in DTs-based smart farming. IoT, UAVs, and AI-enabled predictive models generate massive amounts of data that demand high-performance computing (HPC) resources for processing [96]. However, the architects of cloud-based DTs often introduce a lot of delay due to network latency, delaying real-time decision-making on the farm [97,98]. This problem is of magnified concern in rural agricultural zones where low internet bandwidth and computing power impede instant farm data processing and analysis [99,100]. Due to the high computational complexity of DT models and the network delays, automated farm operations, including resource allocation, machinery control, and crop monitoring, can be interrupted [86,101,102]. One solution to this problem is edge computing, wherein AI models are run on-site rather than solely on a cloud server [103]. This optimization eliminates latency, accelerates data processing, and enhances real-time monitoring, which makes DT-led smart farming seamlessly executable and scalable. Secure data transmission and management are another important concern of DT implementation in agriculture. Blockchain-based security frameworks can help by providing tamper-proof and distributed data governance, traceability, authentication, and protection of farmers' digital assets [104]. These findings will be vital in the scale-up of DT utilization on the farm, enhancing productivity and facilitating more sustainable and smart agricultural systems [86].

6.2. Identifying Key Barriers to DTs Implementation and Exploring Solutions

While DTs are becoming more common in smart farming, several key challenges still prevent their extensive use. One of the biggest issues is data integration, as DT models rely on real-time data from different sources, such as UAVs, IoT sensors, climate stations, and satellite imagery [88]. Since these systems use different data formats and communication protocols, compatibility issues occur, making it difficult to combine and analyze data efficiently. A solution to this problem is the development of universal data exchange protocols and cloud-based platforms that allow different agricultural technologies to work together seamlessly [105,106]. Another major challenge is the high computational power required to run detailed DT simulations. These models require massive processing resources that can be costly and unaffordable for small-scale farmers. Edge computing or distributed cloud processing is one solution, as it can reduce the load on central servers by processing data as close to the source as possible [107]. Scalability is also a challenge since some of the applications from DTs pass through small controlled research settings, and sometimes they fail in large-scale actual farms [108]. To scale these models to larger farms, we need better infrastructure, flexible designs, and adaptive algorithms that can accommodate a wide variety of farm conditions. Additionally, data security and privacy remain a major concern, given that DT systems gather and process vast amounts of sensitive agricultural data, which puts them at risk of cyber-attacks [92].

Addressing these risks requires implementing strong cybersecurity measures, such as the use of encrypted storage and secure data-sharing frameworks [109]. A key obstacle is the absence of technical knowledge and training, especially in developing regions, where farmers and agricultural practitioners may not gain exposure to specialized training in DT technologies [110–112]. Educational program investments and user-friendly DT interfaces can assist in closing this gap, making the technology more accessible [113]. However, the current challenges will not have a major impact on the future development of DTs, as evidenced by the surging growth of DTs research and rapidly increasing industry investment [114]. With continuous advances in DTs for smart farming, this will become more effective and large-scale in the future. Building flexible and scalable frameworks that can help overcome the obstacles of adopting DTs across various farms is the need for the hour, depicted by the different sizes of farms, types of crops, and regional environmental conditions. High implementation costs are one of the greatest barriers, hindering investment in advanced DT infrastructure for small and medium-sized farms. Governments and research organizations can introduce grants and cost-effective cloud-based solutions and provide funds to make DTs more accessible for farmers [115]. Overcoming these constraints through technology development, sound policy, and partnerships in agriculture will enable more farmers to adopt DTs and enhance efficiency, sustainability, and productivity in agriculture [114].

6.3. Comprehensive Digital Twin Framework for Precision Agriculture

The successful integration and optimization of DTs in the context of precision agriculture necessarily require a well-structured framework. This helps ensure that DTs are designed, deployed, and used in such a way as to enhance farm productivity, resource management, and sustainability. The process begins with the Data Acquisition Layer, where real-time data is collected from various sources such as UAVs, IoT sensors, and climate stations [116]. This includes data taken from soil moisture sensors, temperature and humidity monitors, weather forecasting models, and multispectral UAV imagery, which together provide a rich picture of farm conditions [117]. After the data is collected, it progresses to the data processing and fusion layer, in which AI algorithms filter, clean, and integrate information from various sources. This process helps in achieving proper accuracy and

uniformity in the form of the farm ecosystem that is represented digitally. These data points integrated into the modeling and simulation layer apply advanced models to analyze farm conditions and possibly forecast the outcome.

Advanced AI-driven modeling and machine learning gives farmers a prediction of crop growth, disease threats, and irrigation requirements to make informed decisions. The following layer is the Decision-support and Visualization Layer, where the processed data are displayed on user-friendly dashboards and platforms [118]. Such interfaces allow farmers to track real-time farm activities, compare scenarios, and optimize farming strategies. Finally, the farmers use the information from the DTs system to run their business in the implementation and feedback layer. Our system works with raw data until the last minute; it will be processed with the most recent and relevant data available. The structured framework makes DTs one of the most advantageous instruments in precision agriculture, driving farmers towards data-driven decisions, enhancing efficiency, and fostering sustainable agriculture [119]. With the introduction of cutting-edge technologies like IoT, AI, and remote sensing, DTs will offer a real-time and adaptive solution for contemporary farming, resulting in improved resource efficiency and productivity and transforming it into a more robust framework. The first step is the virtual twin development layer, whereby the physical farm environment is mapped to a 3D digital spatial model using GIS-based mapping [120]. Modeling of crop growth using AI and analytics of remote sensing data. This digital farm is kept continuously updated with real-world data and enables simulation and predictive analysis of crop yield, disease outbreaks, and resource consumption [121].

After establishing the virtual model, the DSS Integration Layer translates the insights generated by the DTs into actionable recommendations for farmers [82]. Such as optimized sowing timetables, automated pest management methods, and AI-powered supply chain control to enhance farm productivity. The last layer is the ongoing monitoring and optimization layer, ensuring that the DTs remain dynamic as the environment changes. A real-time feedback loop between the physical and virtual farm, enabling a feedback loop for continuous improvement in farm management. Drawing on our earlier framework, this holistic DTs framework offers a systematic guide to designing, integrating, and optimizing DTs in the context of precision agriculture, ultimately resulting in increased productivity, cost-efficiency, and long-term sustainability.

6.4. Future Research Directions and Policy Implications

Researchers in open-field agriculture have taken considerable steps and are redefining smart farming practices with DT technology [122]. DTs have gained global attention across industries for their innovative concept of creating a virtual copy of the real-world scenario [123]. Despite the rapid growth of these technologies, agricultural DT adoption is still in the early stages, and several challenges must be overcome to fully leverage its potential for large-scale operations [124].

The adoption of modern sensors, UAVs, and IoT-enabled devices within farming systems is still compromised by data standardization, and computational, and large-scale interoperability gaps [125–127]. AI and ML techniques are being leveraged to improve DTs predictive potential, automatic data analysis, and real-time pest control, irrigation scheduling, and fertilization management [128]. One of the most promising opportunities for evolution in DTs-based agriculture is the increasing integration of multi-spectral UAV imaging with AI-powered DTs modeling. Advanced AI solutions can process multispectral and thermal imaging data captured by drones to enable farmers to quickly identify plant stress, detect diseases, and evaluate nutrient levels. Using automated data processing instead of manual field inspection allows us to monitor the farm in a scalable and fast

manner, drastically reducing the time required for it and providing the data in a more accurate manner [129–133]. Moreover, the integration of data from IoT-connected devices, cloud computing, and DTs to make use of real-time data enables instant decisions and improved management of farm resources.

DTs allow farmers to monitor and simulate key agricultural activity such as crop growth, resource use, and logistics, which helps to optimize operations both in the field and throughout the supply chain [134,135]. With predictive analytics, this software receives the right environment parameters and develops a DTs farmer model that fits even without human involvement, and this is only due to the database and information indexed for optimal farm management, including redirecting pests and fertilizer and using the desired amount of water [136]. Moreover, climate-resilient DT modeling is becoming increasingly important as we work to mitigate the impacts of extreme weather patterns, droughts, and soil degradation [137,138].

As climate change emerges as an even greater threat to food security worldwide in the future, we must develop future DTs that not only predict and simulate but also design and optimize farming practices in light of varying environmental conditions [86,139]. Adaptive DT frameworks to react to changes in temperature, precipitation patterns, and soil fertility will play a critical role in ensuring agricultural resilience and sustainability [102,114]. Data security and transparency in smart farming are also the primary challenges, in addition to predictive analytics and climate adaptation. Another layer of complexity that can lend itself well to IoT solutions is blockchain technology, which provides a secure, tamper-proof environment that could enhance the utility of these applications, especially when used in conjunction with DTs to authenticate data reported by devices and even track farm operations, resource consumption, and carbon footprints. Furthermore, cooperative DT platforms enabling cross-farm data exchange can enable numerous farmers and agricultural institutes involved in shared DT models with the aim to enhance overall agricultural expertise and induce cooperative farming strategies. Substantial policy interventions and governmental incentives are needed to enable widespread use of the Digital Twin in agriculture [114,140]. Financial and technical resources to implement sophisticated DT models are out of reach for many small and medium-sized farms [141]. However, governments should support research initiatives, grant subsidies, and make open-access DT platforms available so more farmers can engage in data-driven agriculture. The interoperability of DTs is also going to be important when increasing international agriculture research, trade, and supply chain optimization [142]. Moreover, the future of smart farming technologies will be a combination of AI-enabled DT models, climate-adaptive frameworks, blockchain security, and joint data-sharing platforms. With strategic input on research investment and policy interventions, DTs can potentially drive precision farming on a large scale and ensure a sustainable, technology-empowered agriculture future. Finally, research also needs to address the social and economic impacts of DTs to ensure that rural farmers, agricultural workers, and policymakers can take advantage of these emerging technologies.

Model accuracy and effectiveness are generally assessed using a range of technical, operational, and economic indicators when it comes to DTs in agriculture. Technically, one key measure is prediction accuracy, such as how closely the model can estimate yield, detect diseases, or monitor soil and crop conditions compared to real-world outcomes. Computational feasibility is also important, which looks at whether the DT can process data in real time, how much computing power it needs, and whether it can scale to larger systems. From an operations perspective, researchers investigate efficiency gains, such as more efficient water application rates, reduced fertilizer application rates, or improved labor management. Economic viability is another key aspect, usually evaluated in terms of cost-benefit analysis, return on investment (ROI), or savings stemming from DT-informed

decisions. Currently, the state of the art includes models that combine AI, IoT remote sensing, and in some cases, blockchain and edge computing. These systems often report performance in terms of percentage accuracy (e.g., 85–95% for disease detection) [143], system response time (e.g., seconds for real-time decision-making) [144], and savings (e.g., 20–30% reduction in water use) [145]. However, there is still no universal framework for evaluating DTs in agriculture, and most studies use custom, case-specific metrics. This lack of standardization is a key research gap that future studies should aim to address.

This review highlights the increasing importance of DT technologies in modern agriculture. We found that DTs are being used in many areas such as monitoring crops with drones, managing smart greenhouses, improving irrigation systems, and making better use of resources. Technologies like IoT, artificial intelligence (AI), UAVs, and cyber-physical systems are being combined to support smarter and more efficient farming practices. However, our analysis also showed several research gaps. Most studies are focused on specific use cases and do not provide a complete or standardized framework that can be applied to different types of farms. There is limited information on how to scale these systems for larger farms or different environments. In addition, only a few studies discuss how affordable or practical these technologies are for farmers, especially in low-resource or rural settings.

For future research, there is a need to design flexible DT systems that can be adapted to various farming needs and scales. Researchers should work on improving how different systems and sensors can share and process data more easily. It is also important to test DTs systems in real farm conditions, across different regions, crops, and climates. Lastly, more studies are needed in developing countries where smart agriculture is just beginning to grow but has the potential to make a big difference in food security and sustainability.

7. Conclusions

To conclude, the applications of DT technologies in the domain of precision agriculture and smart farming underline its potential for real-time monitoring, predictive analytics, and automated decision-making. When integrated with the IoT, UAVs, AI, and cloud computing, DTs have the potential to optimize resource utilization, improve crop management, and contribute to sustainable agriculture. Despite growing interest in other industries, the application of DTs in agriculture faces major technological and adaptation challenges, with limited research currently focused on practical use cases. This literature review studies the comprehensive analysis of the adoption of DTs in smart farming and provides future research directions and policy suggestions for the facilitation of its implementation. The findings from this review demonstrate how DTs can support the transition to scalable precision agriculture.

DTs provide meaningful insights into crop health, soil conditions, and climate adaptation methods. This review also highlights the need for self-learning DT models. Traditional DTs use a set of predefined rules to make decisions, restricting their adaptability to changes in the real world. In addition, the development of climate-resilient DT models will aid farmers by enabling adaptation to extreme weather conditions, droughts, and soil degradation. They will also be key in ensuring food security and sustainable farming practices and mitigating climate change. Moreover, collaborative DT platforms, in which several farmers exchange real-time data, can help increase collective intelligence, optimize supply chain efficiency, and minimize agricultural risks. From a policy perspective, governments and research organizations will have to play an active role in promoting the uptake of DTs in agriculture. Subsidies, research funding, and open-access platforms will facilitate the widespread adoption of smart farming technologies. Given the right combination of strategic investments, technological innovations, and policy support, DTs will position

agriculture as an industry of the future characterized by high efficiency, resilience, and data-driven decision-making to enable long-term productivity and environmental sustainability.

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